
Notes for ‘The Product Rule’

Important Idea and Useful Fact:

The Product Rule: Suppose that $y = uv$ where u and v are differentiable functions of x . Then, using Leibniz notation,

$$\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx} ,$$

or, equivalently, using function notation,

$$y' = uv' + vu' .$$

Examples and derivations:

1. Differentiate $y = (3x^2 + 6x - 7)(4x + 1)$.

Solution (direct): Expanding the brackets, we have

$$y = 12x^3 + 3x^2 + 24x^2 + 6x - 28x - 7 = 12x^3 + 27x^2 - 22x - 7 ,$$

so that

$$y' = 36x^2 + 54x - 22 .$$

Solution (using the Product Rule): We have $y = uv$ where $u = 3x^2 + 6x - 7$, so that $u' = 6x + 6$, and $v = 4x + 1$, so that $v' = 4$. By the Product Rule,

$$\begin{aligned} y' &= uv' + vu' = (3x^2 + 6x - 7)(4) + (4x + 1)(6x + 6) \\ &= 12x^2 + 24x - 28 + 24x^2 + 24x + 6x + 6 = 36x^2 + 54x - 22 , \end{aligned}$$

which, of course, agrees with the direct solution.

2. Differentiate $y = (3 - e^{2x})(2 + e^{-x} - e^{5x})$.

Solution (direct): Expanding the brackets, and using an exponential law, we have

$$\begin{aligned} y &= 6 + 3e^{-x} - 3e^{5x} - 2e^{2x} - e^{2x}e^{-x} + e^{2x}e^{5x} \\ &= 6 + 3e^{-x} - 3e^{5x} - 2e^{2x} - e^x + e^{7x} , \end{aligned}$$

so that, using the fact that $\frac{d}{dx}(e^{kx}) = ke^{kx}$ for any constant k ,

$$y' = -3e^{-x} - 15e^{5x} - 4e^{2x} - e^x + 7e^{7x} .$$

Solution (using the Product Rule): We have $y = uv$ where $u = 3 - e^{2x}$, so that $u' = -2e^{2x}$, and $v = 2 + e^{-x} - e^{5x}$, so that $v' = -e^{-x} - 5e^{5x}$. By the Product Rule,

$$\begin{aligned} y' &= uv' + vu' = (3 - e^{2x})(-e^{-x} - 5e^{5x}) + (2 + e^{-x} - e^{5x})(-2e^{2x}) \\ &= -3e^{-x} - 15e^{5x} + e^{2x}e^{-x} + 5e^{2x}e^{5x} - 4e^{2x} - 2e^{-x}e^{2x} + 2e^{5x}e^{2x} \\ &= -3e^{-x} - 15e^{5x} + e^x + 5e^{7x} - 4e^{2x} - 2e^x + 2e^{7x} \\ &= -3e^{-x} - 15e^{5x} - e^x + 7e^{7x} - 4e^{2x} , \end{aligned}$$

which, of course, agrees with the direct solution.

3. Find $\frac{dy}{dt}$ when $y = t^2 e^{-3t}$.

Solution: We have $y = uv$ where $u = t^2$ and $v = e^{-3t}$, so that $\frac{du}{dt} = 2t$ and $\frac{dv}{dt} = -3e^{-3t}$. By the Product Rule,

$$\frac{dy}{dt} = u \frac{dv}{dt} + v \frac{du}{dt} = t^2(-3e^{-3t}) + e^{-3t}(2t) = te^{-3t}(2 - 3t).$$

4. Find y' when $y = x^3 \cos(2x)$.

Solution: We have $y = uv$ where $u = x^3$, so that $u' = 3x^2$, and $v = \cos(2x)$, so that $v' = -2\sin(2x)$, by the Chain Rule. By the Product Rule,

$$y' = uv' + vu' = x^3(-2\sin(2x)) + \cos(2x)(3x^2) = x^2(3\cos(2x) - 2x\sin(2x)).$$

5. Find $\frac{du}{dt}$ when $u = t^2 \ln(2t)$.

Solution: We have $u = vw$ where $v = t^2$ and $w = \ln(2t)$. Then $\frac{dv}{dt} = 2t$ and, applying the Chain Rule, noting a common natural variation in the notation, just to avoid introducing a new symbol in making the substitution,

$$\frac{dw}{dt} = \frac{d(\ln(2t))}{d(2t)} \frac{d(2t)}{dt} = \left(\frac{1}{2t}\right)(2) = \frac{2}{2t} = \frac{1}{t}.$$

By the Product Rule,

$$\frac{du}{dt} = v \frac{dw}{dt} + w \frac{dv}{dt} = t^2\left(\frac{1}{t}\right) + (\ln(2t))(2t) = t(1 + 2\ln(2t)).$$

6. Find $\frac{dx}{d\theta}$ when $x = \sin^5 \theta \cos^4 \theta$.

Solution: By the Product and Chain Rules, we have

$$\begin{aligned} \frac{dx}{d\theta} &= (\sin^5 \theta) \left(\frac{d}{d\theta} (\cos^4 \theta) \right) + (\cos^4 \theta) \left(\frac{d}{d\theta} (\sin^5 \theta) \right) \\ &= (\sin^5 \theta) \left(\frac{d(\cos^4 \theta)}{d(\cos \theta)} \frac{d(\cos \theta)}{d\theta} \right) + (\cos^4 \theta) \left(\frac{d(\sin^5 \theta)}{d(\sin \theta)} \frac{d(\sin \theta)}{d\theta} \right) \\ &= (\sin^5 \theta) (4 \cos^3 \theta) (-\sin \theta) + (\cos^4 \theta) (5 \sin^4 \theta) (\cos \theta) \\ &= \sin^4 \theta \cos^3 \theta (5 \cos^2 \theta - 4 \sin^2 \theta). \end{aligned}$$

7. Find y' when $y = e^x \sin x \cos x$.

Solution: By the Product Rule,

$$\begin{aligned} y' &= e^x (\sin x \cos x)' + \sin x \cos x (e^x)' \\ &= e^x \left((\sin x) (\cos x)' + (\cos x) (\sin x)' \right) + \sin x \cos x (e^x) \\ &= e^x \left((\sin x) (-\sin x) + (\cos x) (\cos x) \right) + e^x \sin x \cos x \\ &= e^x (\sin x \cos x + \cos^2 x - \sin^2 x). \end{aligned}$$

8. We use Leibniz notation to sketch a proof of the Product Rule. Suppose $y = uv$ where u and v are functions of x . We claim $y' = uv' + vu'$.

Sketch of proof: Consider small changes in x , denoted by Δx , which propagate small changes Δy , Δu and Δv in y , u and v respectively. By this, we mean that

$$\Delta y = y(x + \Delta x) - y(x), \quad \Delta u = u(x + \Delta x) - u(x) \quad \text{and} \quad \Delta v = v(x + \Delta x) - v(x) .$$

We have the following (at the second line, by an insertion of a carefully chosen expression, together with its negative, which does not affect the overall value of the entire expression):

$$\begin{aligned} \Delta y &= y(x + \Delta x) - y(x) = u(x + \Delta x)v(x + \Delta x) - u(x)v(x) \\ &= u(x + \Delta x)v(x + \Delta x) - u(x + \Delta x)v(x) + u(x + \Delta x)v(x) - u(x)v(x) \\ &= u(x + \Delta x)\left(v(x + \Delta x) - v(x)\right) + \left(u(x + \Delta x) - u(x)\right)v(x) \\ &= u(x + \Delta x)\left(v(x + \Delta x) - v(x)\right) + v(x)\left(u(x + \Delta x) - u(x)\right) \\ &= u(x + \Delta x)\Delta v + v(x)\Delta u . \end{aligned}$$

Hence, by the limit definition of the derivative, using Leibniz notation, and making free use of limit laws,

$$\begin{aligned} \frac{dy}{dx} &= \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{u(x + \Delta x)\Delta v + v(x)\Delta u}{\Delta x} \\ &= \left(\lim_{\Delta x \rightarrow 0} \frac{u(x + \Delta x)\Delta v}{\Delta x} \right) + \left(\lim_{\Delta x \rightarrow 0} \frac{v(x)\Delta u}{\Delta x} \right) \\ &= \left(\lim_{\Delta x \rightarrow 0} u(x + \Delta x) \right) \left(\lim_{\Delta x \rightarrow 0} \frac{\Delta v}{\Delta x} \right) + v(x) \left(\lim_{\Delta x \rightarrow 0} \frac{\Delta u}{\Delta x} \right) \\ &= u(x) \frac{dv}{dx} + v(x) \frac{du}{dx} , \end{aligned}$$

(using continuity of u at the last step). This completes the sketch of the proof.